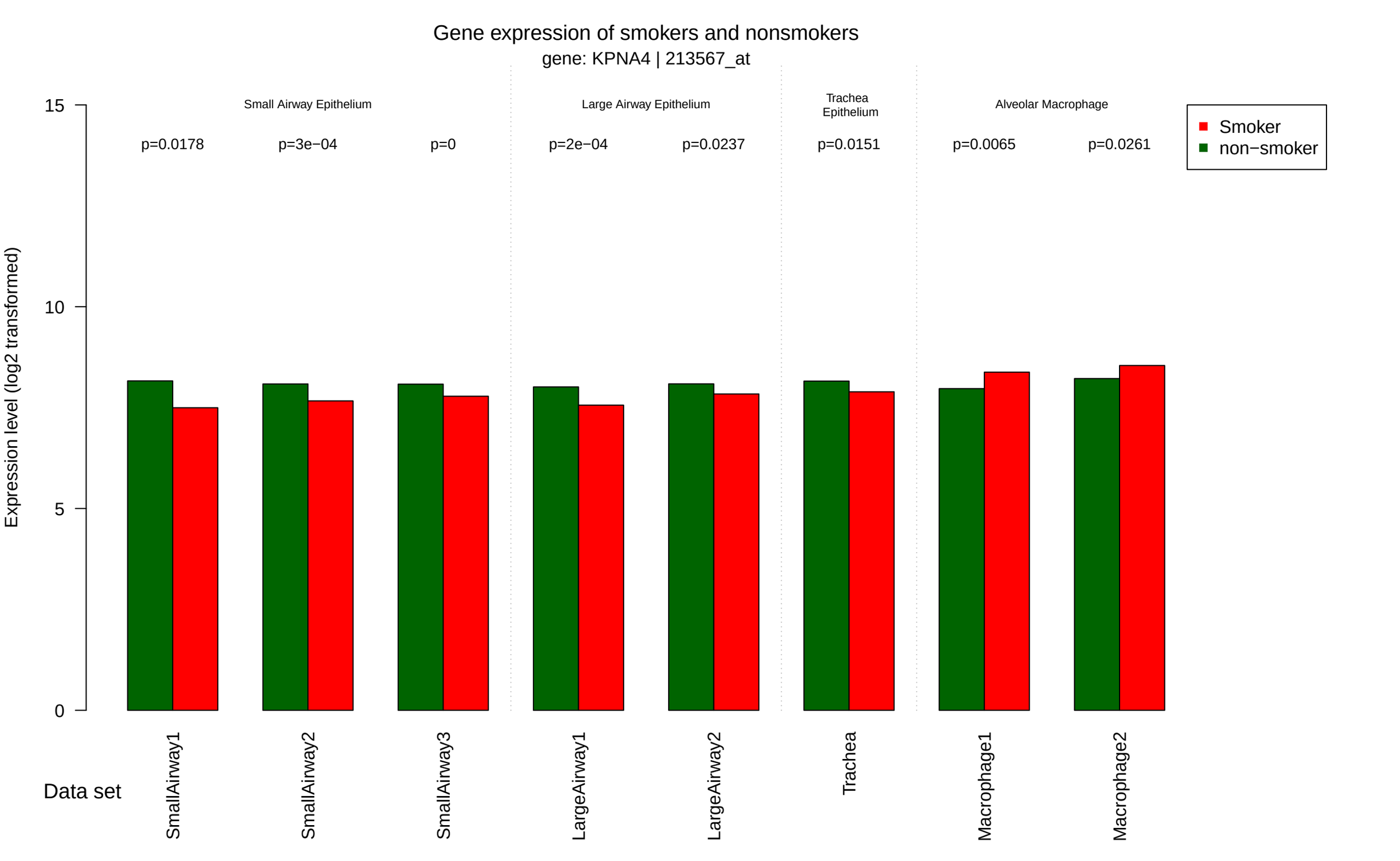
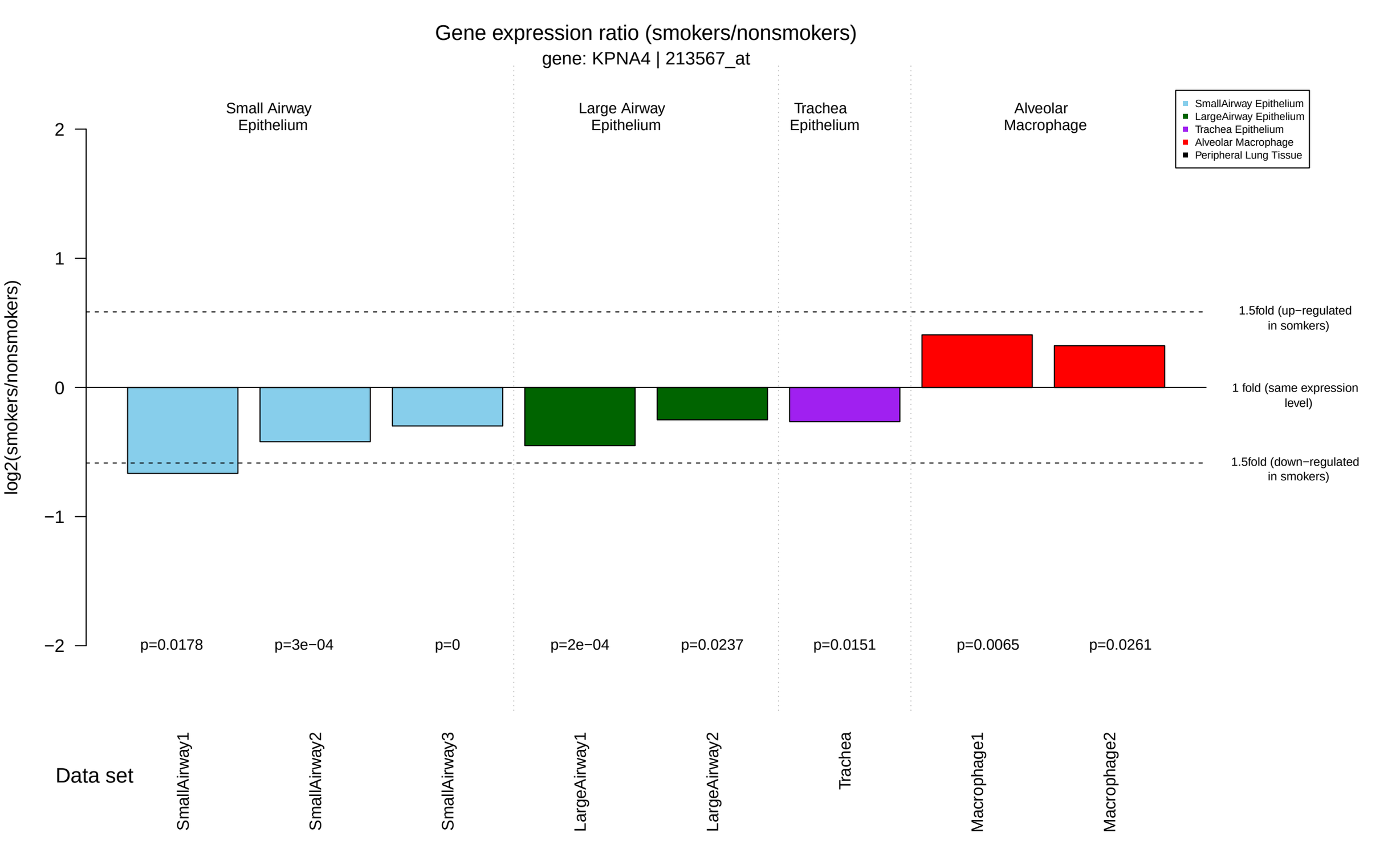
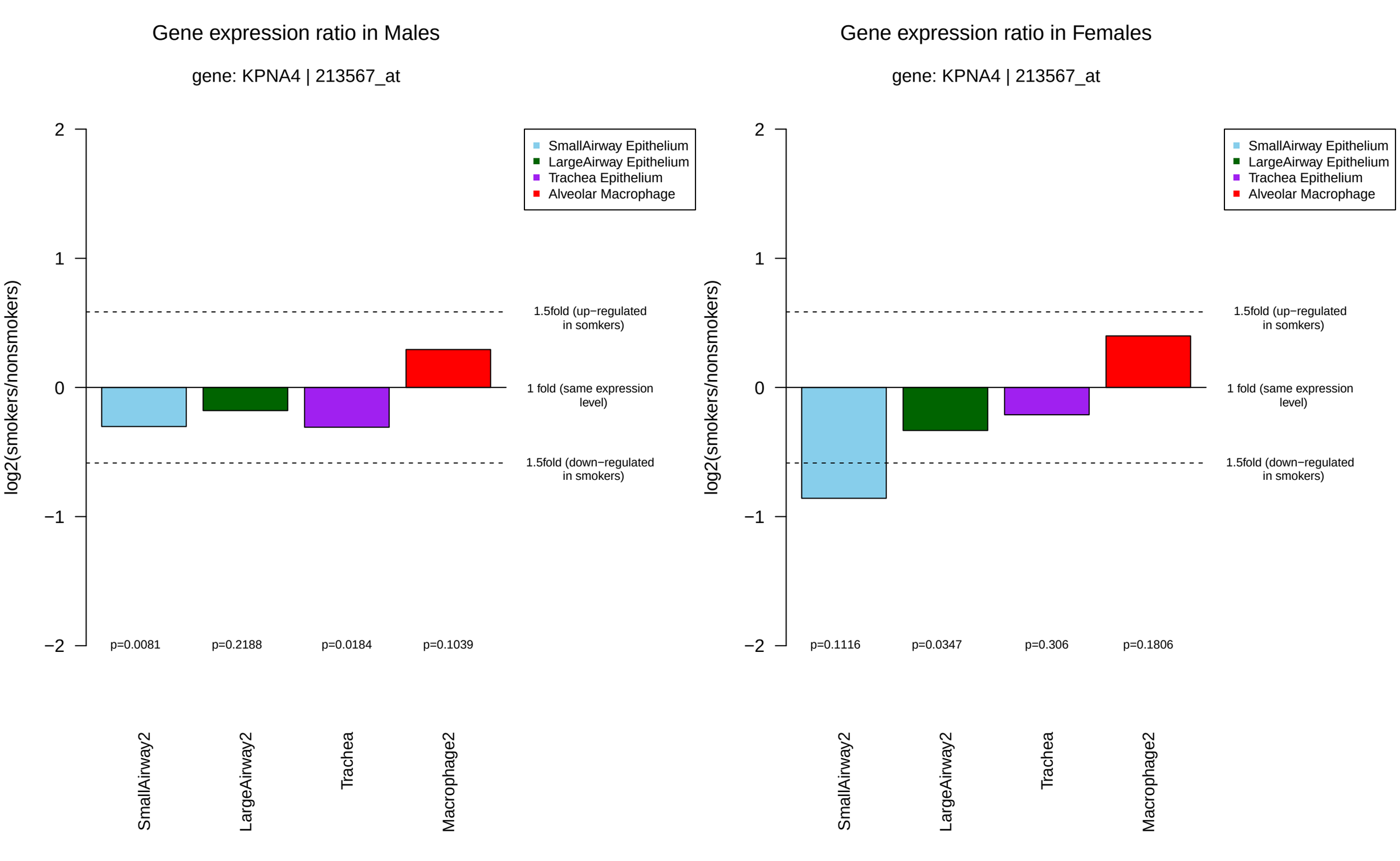
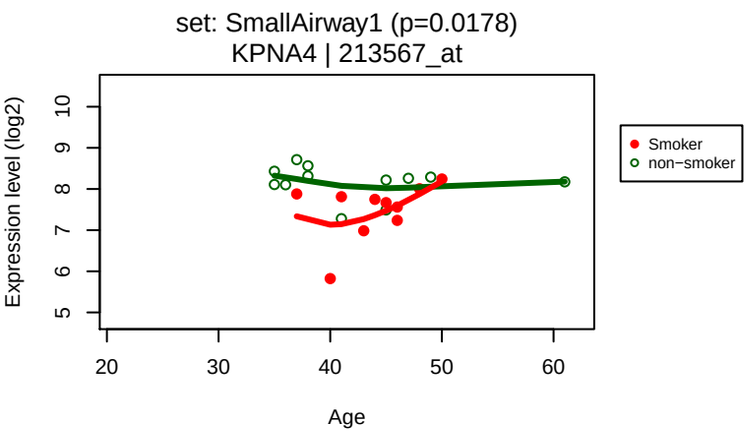
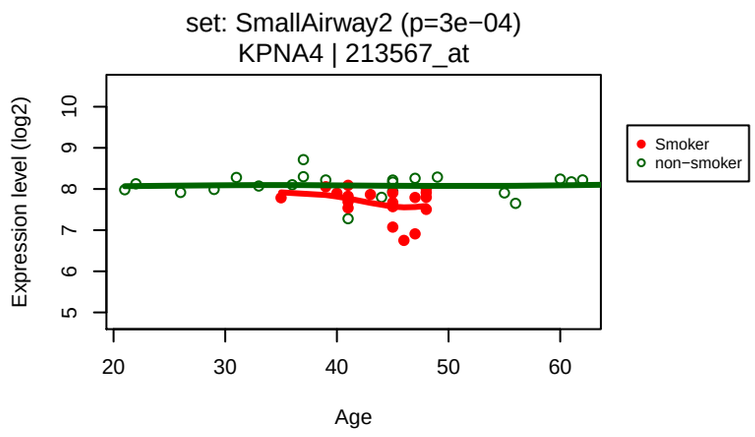
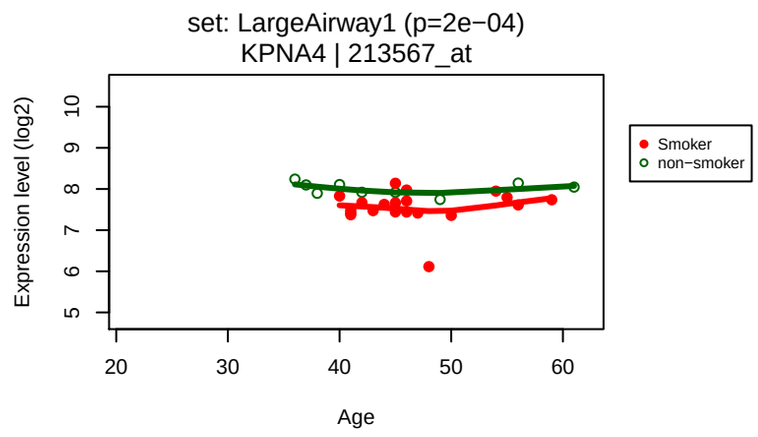

Gene expression of smokers and nonsmokers
gene: KPNA4 | 213567_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0178
p=3e−04
p=0
p=2e−04
p=0.0237
p=0.0151
p=0.0065
p=0.0261
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

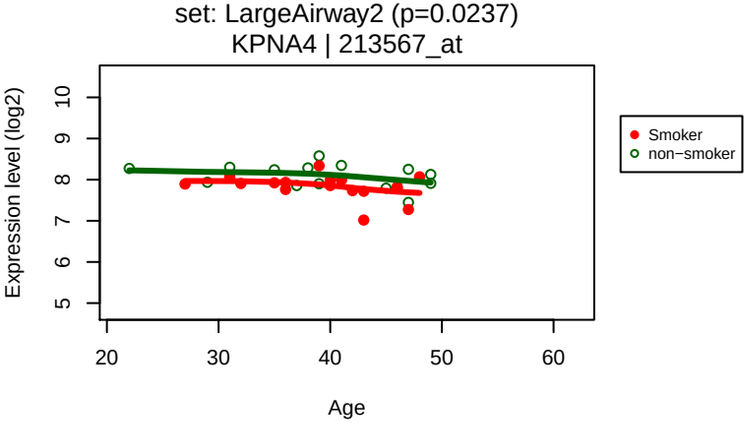

Gene expression ratio (smokers/nonsmokers)
gene: KPNA4 | 213567_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0178
p=3e−04
p=0
p=2e−04
p=0.0237
p=0.0151
p=0.0065
p=0.0261
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

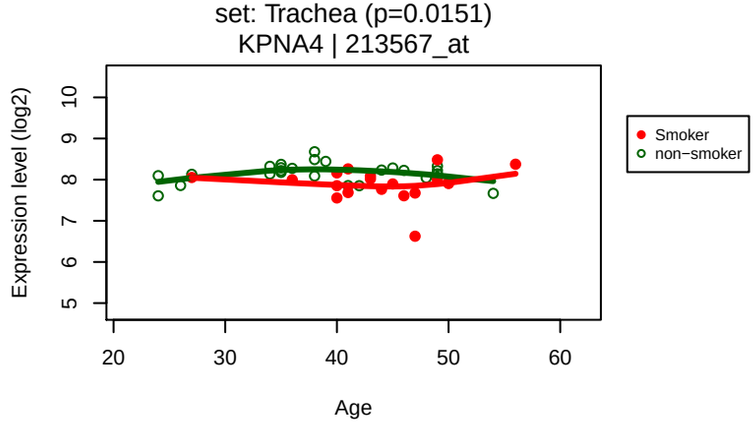

Gene expression ratio in Males
gene: KPNA4 | 213567_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0081
p=0.2188
p=0.0184
p=0.1039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA4 | 213567_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1116
p=0.0347
p=0.306
p=0.1806
SmallAirway2
LargeAirway2
Trachea
Macrophage2

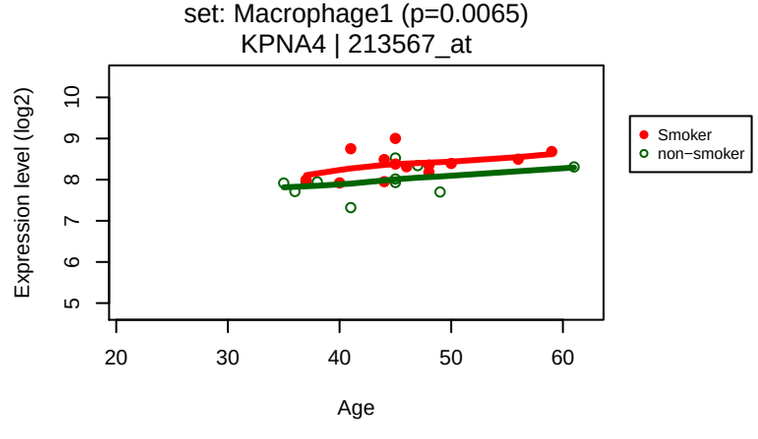

set: SmallAirway1 (p=0.0178)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker

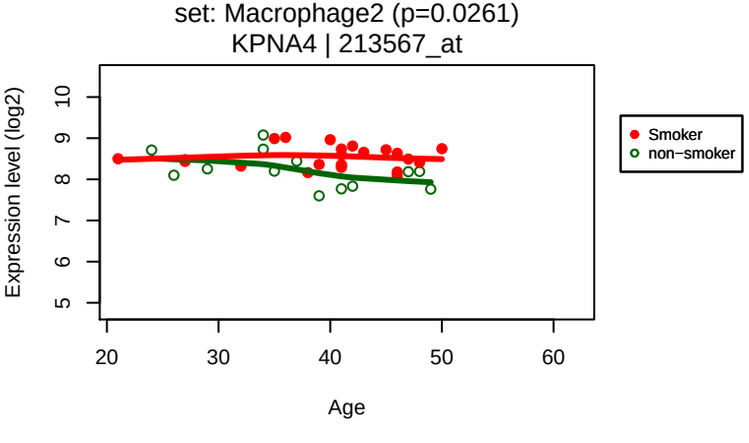

set: SmallAirway2 (p=3e−04)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker

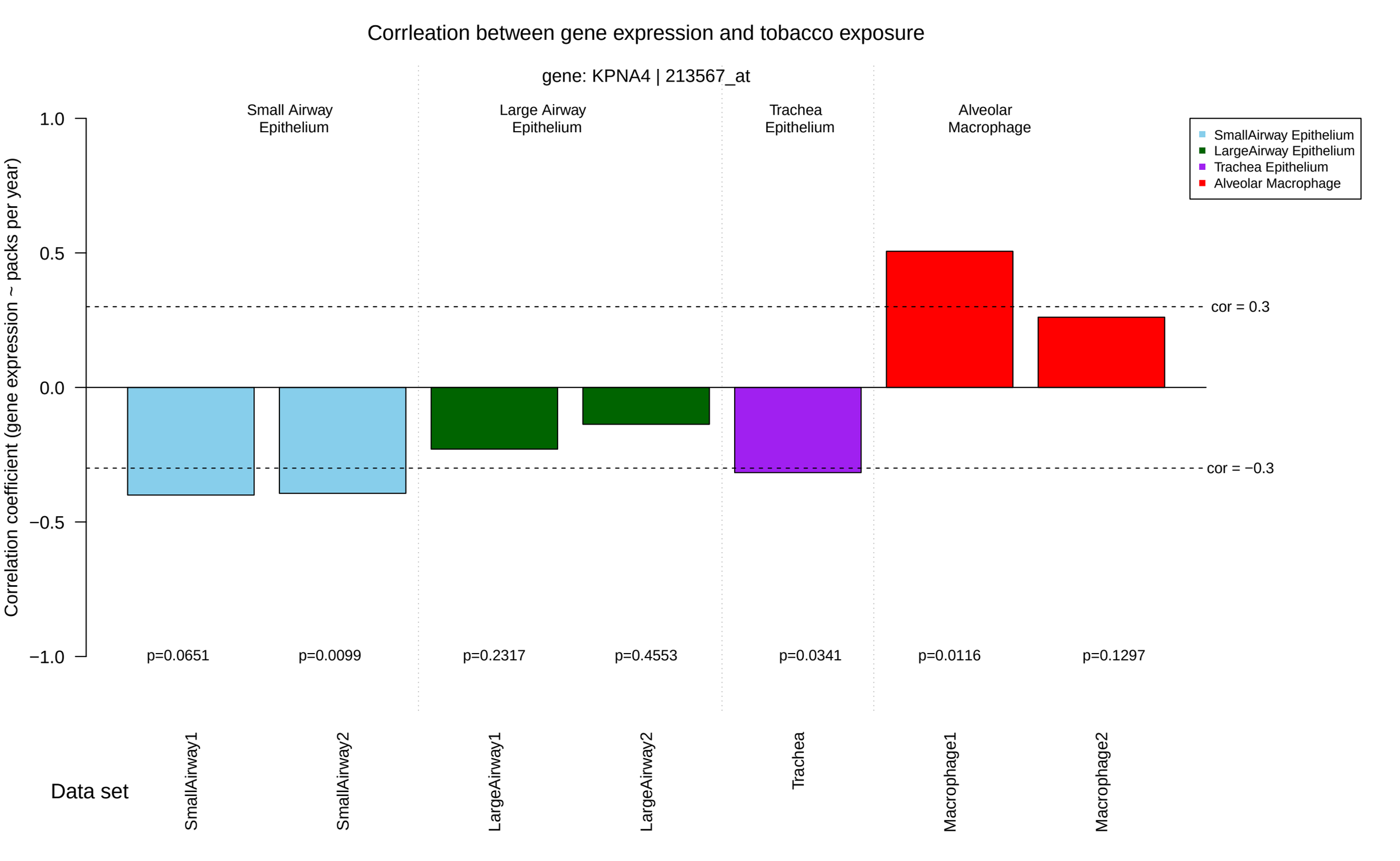

set: LargeAirway1 (p=2e−04)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0237)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0151)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0065)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0261)
KPNA4 | 213567_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA4 | 213567_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0651
p=0.0099
p=0.2317
p=0.4553
p=0.0341
p=0.0116
p=0.1297
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2